

```

clc; clear; close all;

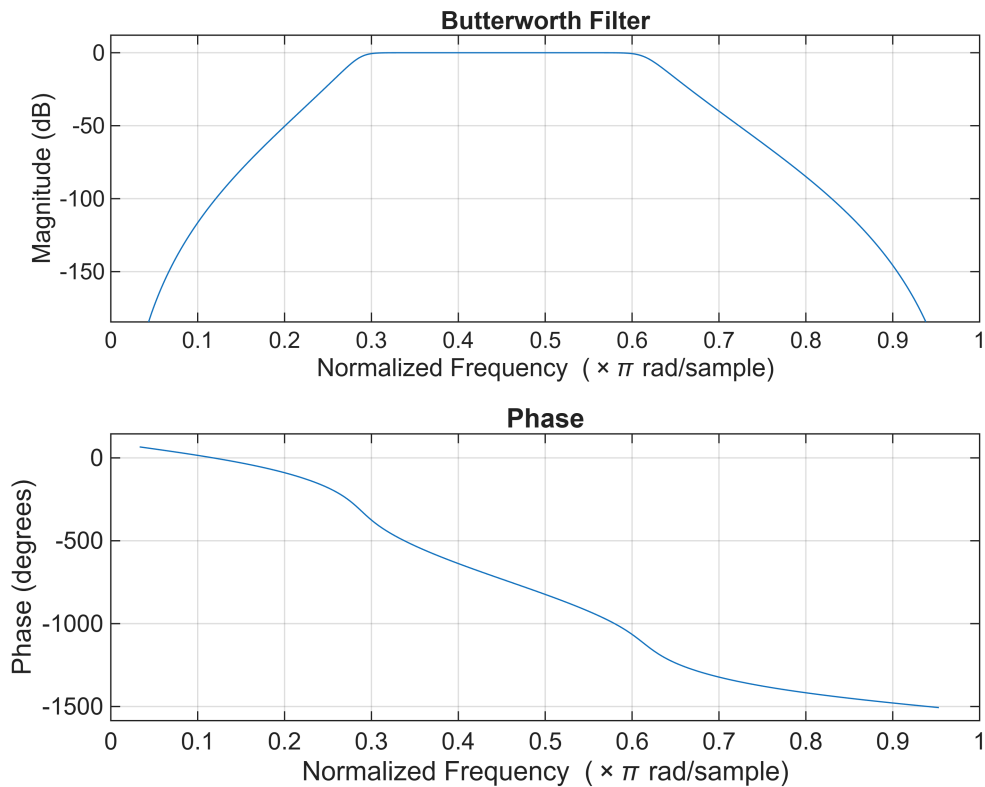
% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');

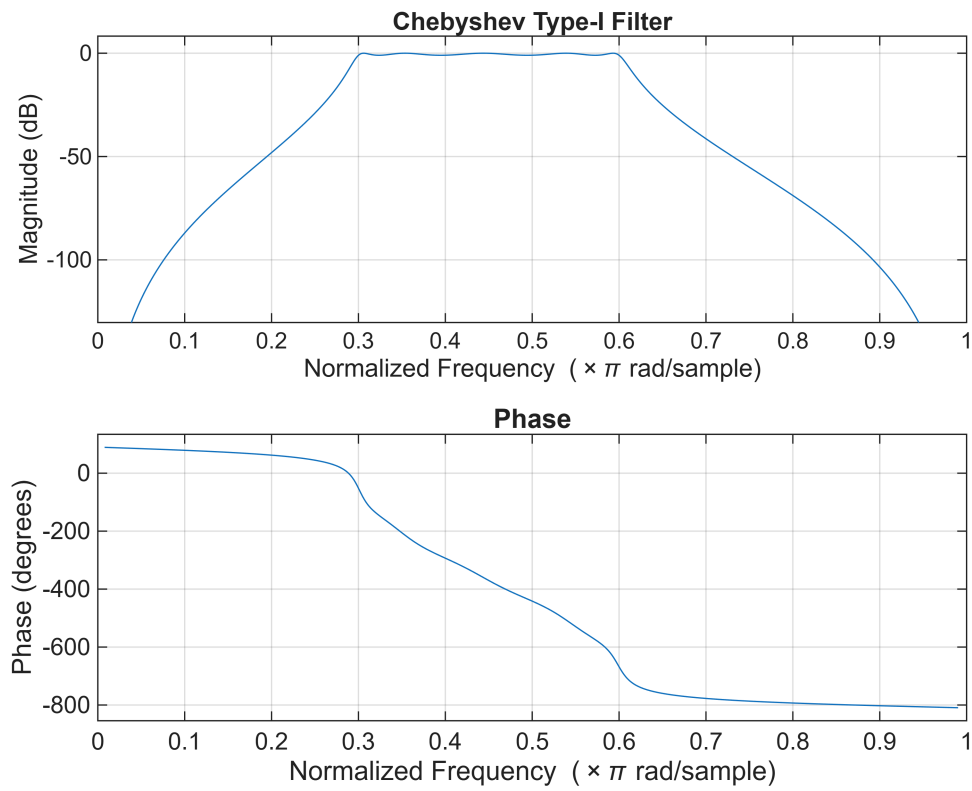
```



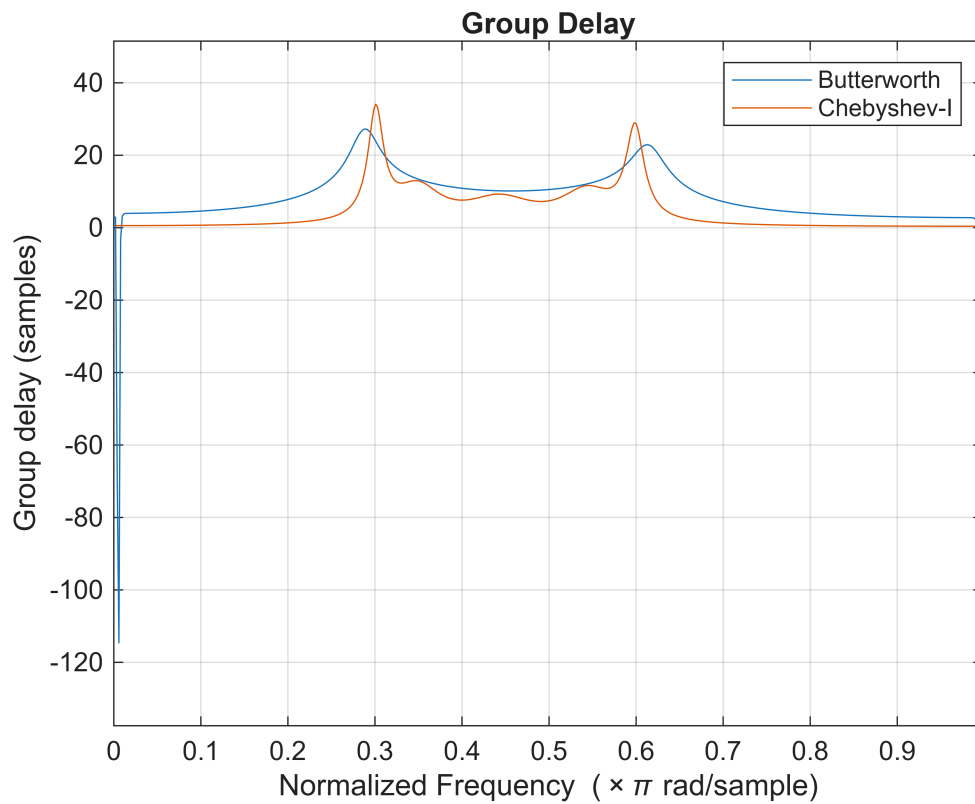
```

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

```

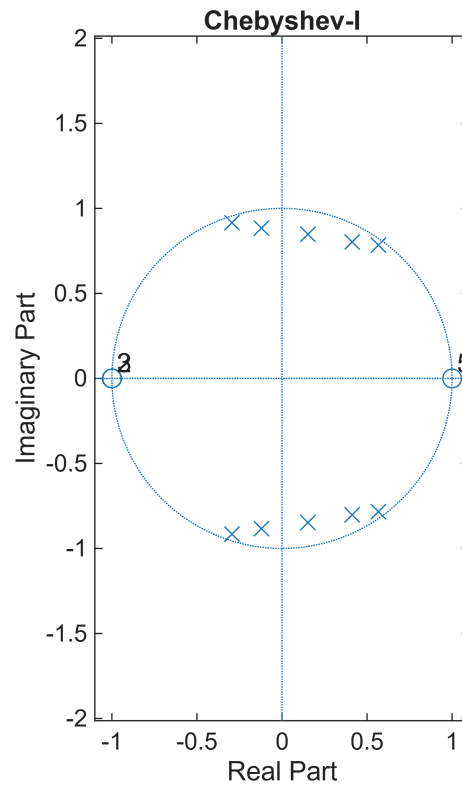
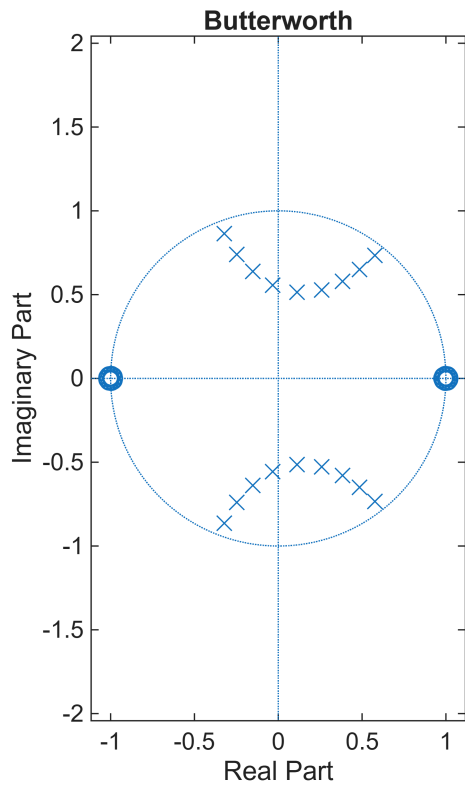


```
% Group Delay  
figure;  
grpdelay(b1,a1); hold on;  
grpdelay(b2,a2);  
legend('Butterworth', 'Chebyshev-I');
```



% Pole-Zero Plot

```
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')
```

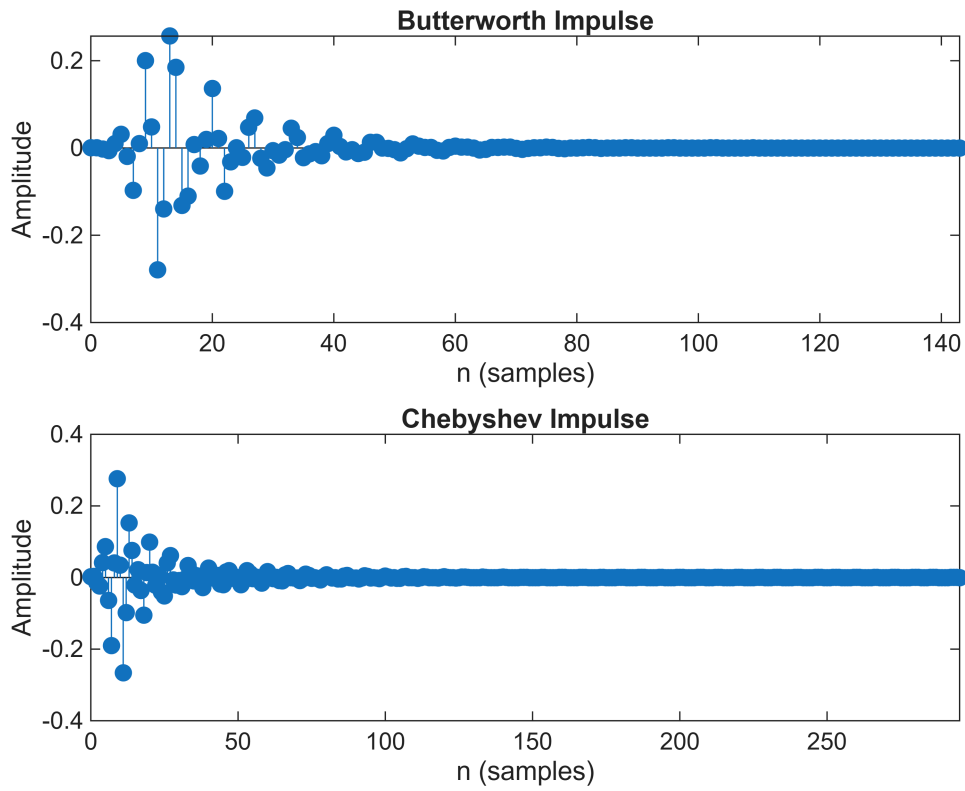


% Impulse Response

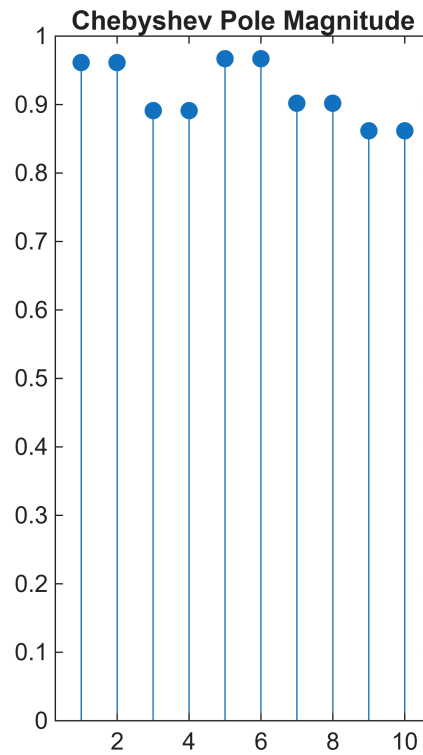
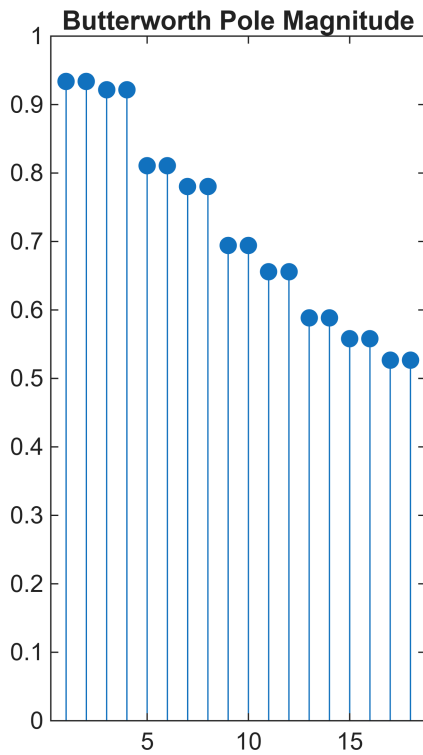
figure;

subplot(211),impz(b1,a1),title('Butterworth Impulse')

subplot(212),impz(b2,a2),title('Chebyshev Impulse')



```
% Pole Magnitude
figure;
subplot(121),stem(abs(roots(a1)), 'filled')
title('Butterworth Pole Magnitude')
subplot(122),stem(abs(roots(a2)), 'filled')
title('Chebyshev Pole Magnitude')
```



```
disp(['Butterworth Order = ',num2str(n1)])
```

Butterworth Order = 9

```
disp(['Chebyshev Order = ',num2str(n2)])
```

Chebyshev Order = 5